

Personal 3D Printing without Visual Dependency in the Generative AI Era: Opportunities and Challenges

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Abstract

Personal fabrication ecosystems of 3D design and desktop 3D printing are largely structured around visual interaction, limiting meaningful access for **Blind and Low-Vision (BLV)** users. This work presents first-person observations in which BLV researchers engaged in an exploratory investigation into **AI-mediated personal fabrication system flow** with text-guided 3D generation, conversational inspection, and artifact printing.

Through implementing and experiencing this AI-integrated workflow, we identify limitations in **dimension control**, **reference-based editing**, and **post-print interpretation** when relying on such AI-driven features. This work outlines **preliminary design implications** to surface challenges in relying on general-purpose AI functionality while providing a formative grounding for future research.

AI-Mediated System Workflow

This section details the end-to-end pipeline enabling BLV users to transition from initial text prompts to physical artifacts using a suite of AI-driven generation and inspection tools.

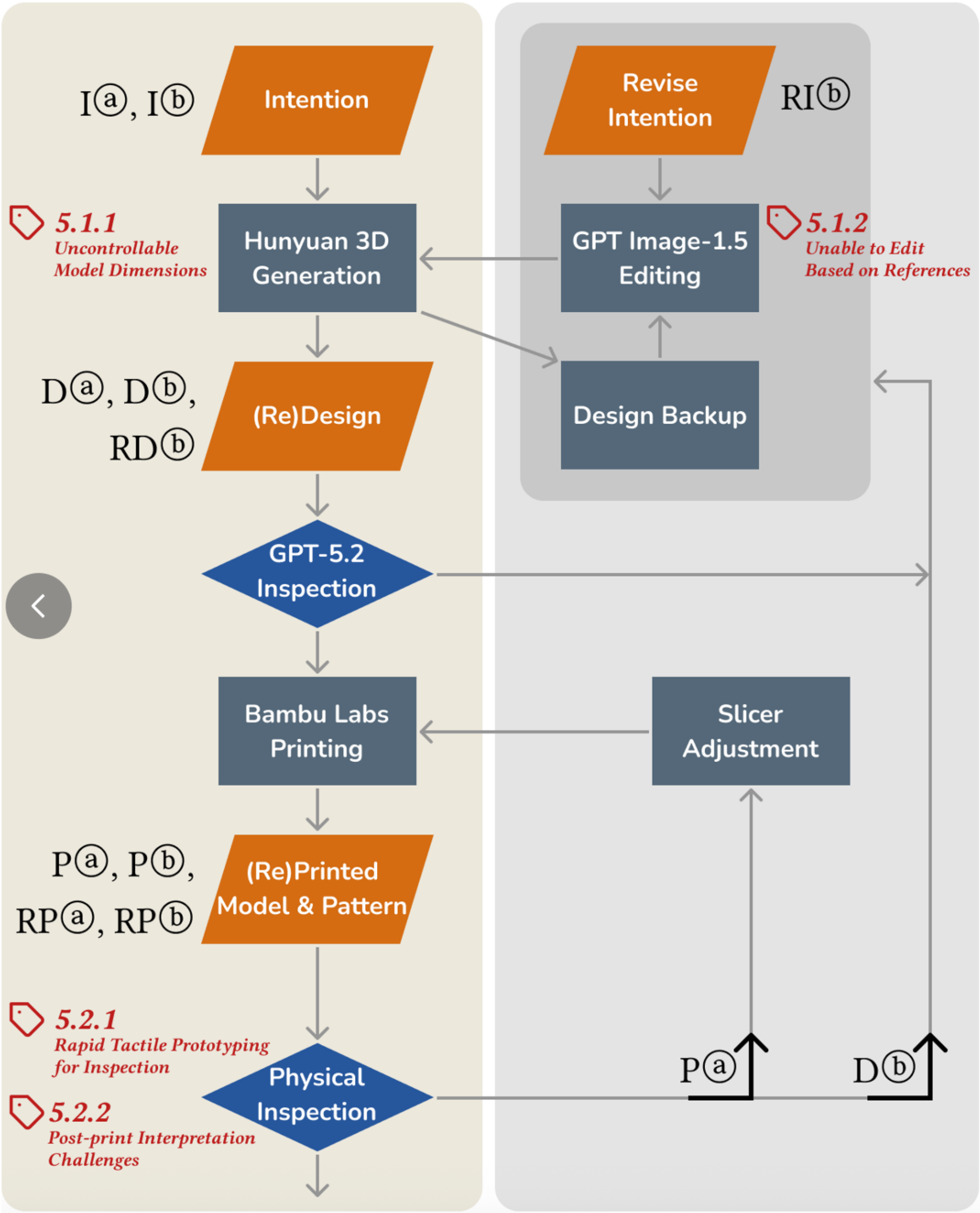


Figure 1: Overview of the end-to-end AI fabrication workflow.

Core Architectural Components:

- **3D Generation:** Leveraged Hunyuan3D 2.5 for text-to-3D generation.
- **Conversational Inspection:** Built a multimodal chatbot via GPT-5.2 to inspect 3D clips.
- **Detour Editing Workflow:** Integrated GPT Image-1.5 for semantic image modifications to bypass lack of direct 3D mesh editing tools.
- **Accessible Front-End:** HTML framework utilizing the Web Speech API for real-time text-to-speech support.
- **Parallel Inspection Paths:** Programmed a custom code generator producing *flat tactile orthographic prints* (i.e., *silhouettes of three main views*) to reduce material overhead.

Exploratory Design Process

BLV researchers explored two distinctive design-fabrication tracks using the developed system framework, evaluating their digital-to-physical transitions.



Figure 2: Digital previews of the exploratory design artifacts.

Flow (A): Game Mascot Design

- **Intent:** Mascot character for a blind-accessible 3D shooting game.
- **Inspection:** 5 minutes of conversation.
- **Fabrication:** Bambu Lab ecosystem.
- **Tactile Preview:** Evaluated via the lightweight *silhouette tactile pattern*.
- **Iteration:** Minor slicer adjustment required before final re-printing.

Flow (B): Tactile Keychain Design

- **Intent:** Maine Coon cat keychain with dimensional constraints.
- **Inspection:** 5 minutes of conversation.
- **Fabrication:** Bambu Lab ecosystem.
- **Tactile Preview:** Evaluated via the lightweight *silhouette tactile pattern*.
- **Detour Editing:** Image-based adjustment to remove outer rim and match waterfall displays.

Evaluation and Core Fabrication Insights

Core Empirical Contributions:

This section presents our findings of the AI-assisted fabrication workflow. Although generative AI enabled rapid creation of 3D designs, several challenges appeared in the physical fabrication process that were not apparent during conversational design and inspection. We discuss observations related to dimension control, rapid prototyping, and post-print ambiguity, and use these experiences to highlight design opportunities for future non-visual personal fabrication systems.

Dimension Control Limitations

- **Reliability Failures:** Quantitative attributes are overlooked by text-guided systems.
- **Mascot $P(A)$ Scale:** Size omitted during generation, yielding an un-usable 12 mm print.
- **Keychain $P(B)$ Bias:** Modeled under explicit multi-dimension constraints but printed at 28 mm instead of 40 mm.
- **BLV Danger:** Issues remain invisible until printing; requires early dimension-aware feedback.

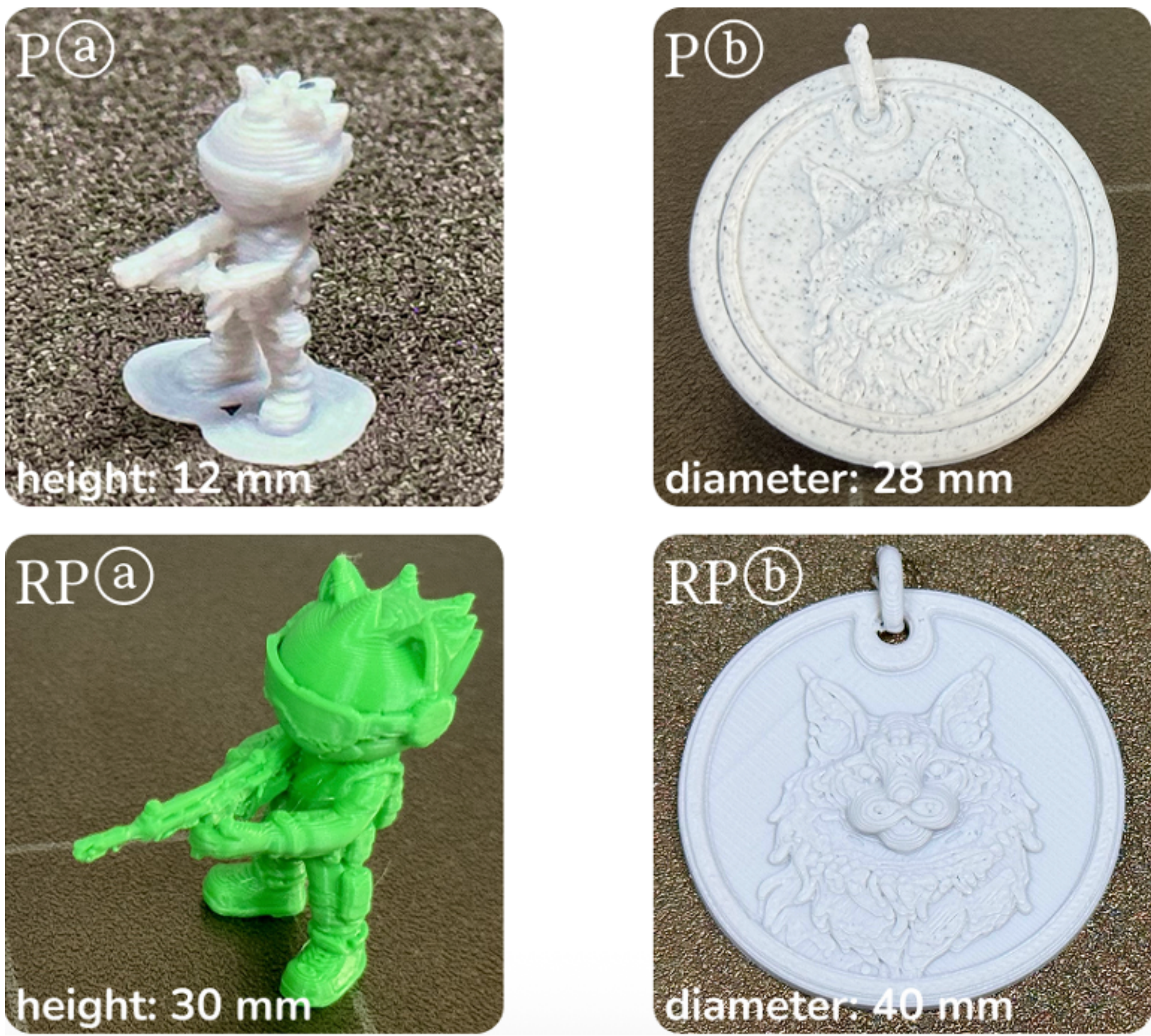


Figure 3: Scale discrepancies discovered during physical artifact inspection.

Rapid Tactile Prototyping for Inspection

- **Immediacy Benefit:** Tactile silhouette patterns for $RP(A)$ and $RP(B)$ print under 10 mins (vs 40/28 mins).
- **Information Trade-off:** Fast prints provide coarse scale data but lack exploratory depth.
- **Strategy Setup:** For small tasks, full prints dominate. For long tasks, scaled-down approximations serve well as lightweight previews.

Post-Print Ambiguities and Support Structures

- **Overhang Deformation:** Printing $P(A)$ without support caused structural failures indistinguishable from design intents.
- **Tactile Blending:** On $RP(A)$, tree support by-products were hard to distinguish from the functional geometric features by touch.
- **Inaccessibility:** Residual support material remained trapped in tight, finger-inaccessible recessed spaces.

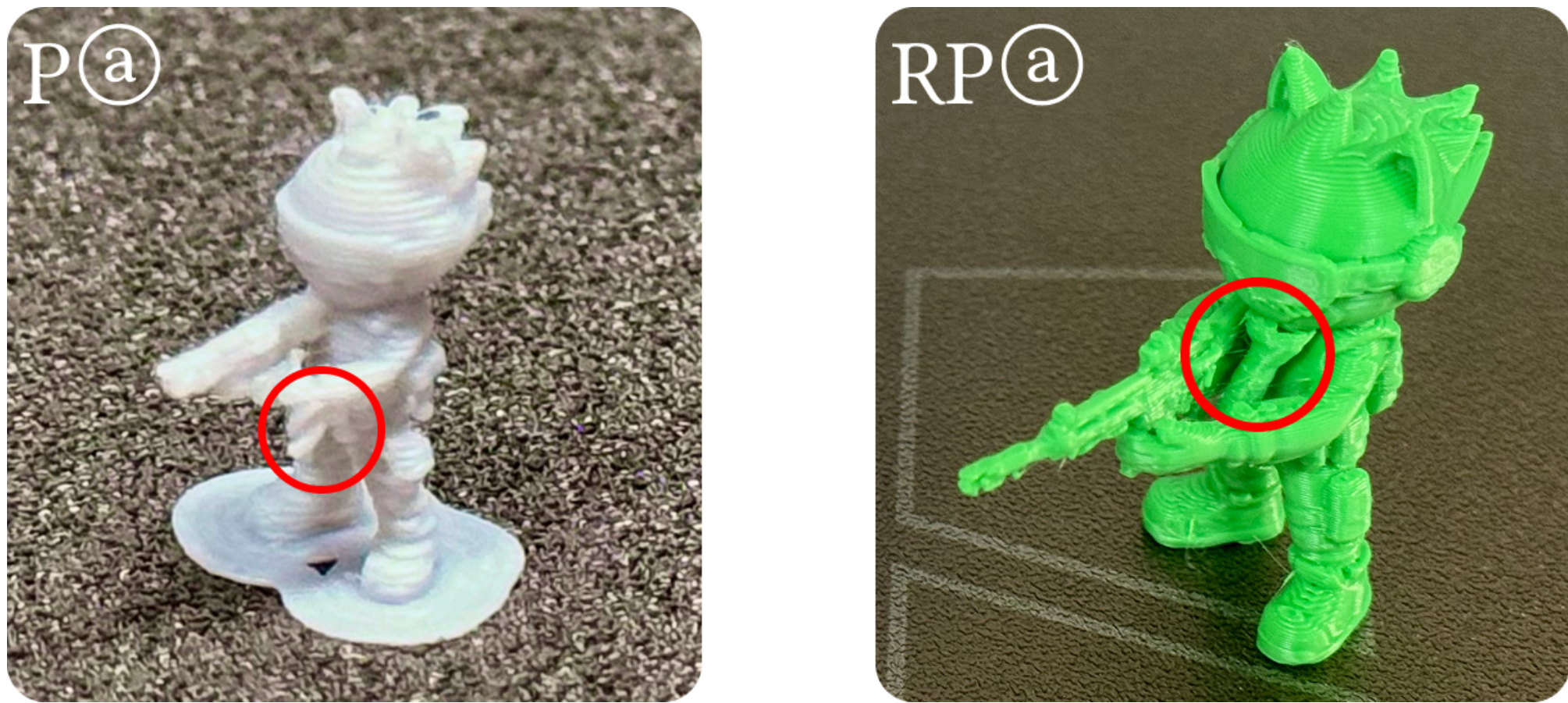


Figure 4: Structural print failures (left) and residual support material (right).

Conclusion and Resources

- Balancing generation velocity with precise physical accuracy remains the primary barrier to autonomous fabrication for BLV populations.
- Future paradigms must shift beyond basic visual prompting toward spatial reasoning engines and tactile-distinct structural slicing algorithms.
- We thank Fab City Challenge High School program for the platform of conducting this project. Please scan the QR code and look the contributions by individual high school students of our team.

